A Named Entity Recognition Model for Threat Intelligence Based on Similar Semantic Space Construction

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Abstract

Extracting cybersecurity entities from unstructured web text is an important part of security analysis. Traditional BERT-based entity extraction models rely on a single semantic expression generated by the model, however, these models do not take into account the polysemous information that exists for entities in the intelligence domain, thus limiting the performance of traditional models for extracting entities. Therefore, this paper proposes a Semantic Enhancement Model (SEM) for named entities based on similar semantic space construction. First, the input text goes into BERT and CNN to get word-level word embeddings and character-level word embeddings respectively, and then merges them into mixed feature word embedding to enhance the robustness of the model. At the same time, SEM computes the set of similar words S-SET for the input text sequentially by the Similar Word Sense Matching Algorithm, and then uses the self-attention mechanism to compute the semantic relevance weights among the S-SET elements, so as to get the word embeddings with similar semantics. Subsequently mixed feature word embeddings and similar semantic word embeddings are used to calculate the similarity between the embedding vectors at different semantic levels through a multi-head attention mechanism, and then fused to construct an integrated semantic expression of the entity; thereby enriching the semantic expression of the entity and increasing the probability of the entity being predicted correctly. Finally, the integrated semantic expression is input into CRF for entity prediction. The experimental outcomes demonstrate that our proposed model outperforms existing baseline approaches on the network security datasets DNRTI and Bridge.

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CCS Concepts

• **Applied computing** → Document management and text processing; Document capture; Document analysis.

Keywords

Threat intelligence, Named entity, Semantic enhancement, Integrated semantics

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1 INTRODUCTION

The rapid advancement of cyber technology has led to the emergence of various cyber threats in everyday life, resulting in the generation of substantial threat intelligence data. This intelligence is typically consolidated into reports, such as security bulletins, Common Vulnerabilities and Exposures (CVE) reports, and National Vulnerability Database (NVD) reports.

However these reports are often presented as unstructured and heterogeneous text from diverse sources, complicating the understanding of cyberattacks and the decision-making process regarding risks. With advancements in deep learning, security analysts have increasingly employed various natural language processing techniques to extract and analyze data related to cyberattacks. Consequently, Named Entity Recognition, Relationship Extraction, and Text Summarization have emerged as prominent research areas. Notably, entity extraction from threat intelligence is a crucial step in this analytical process.

This study focuses on the Named Entity Recognition (NER) task. In recent years, NER research within the context of Cyber Threat Intelligence (CTI) has garnered significant attention, and these studies can be classified into three categories: rule-based approaches, statistical feature-based methods [1], and deep learning-based techniques [2].

Although existing neural models have achieved remarkable success, the following challenges remain. Due to the fact that entities in the cybersecurity domain exhibit polysemous properties, which leads to more complex semantics of intelligence data, resulting in inefficient extraction of entities in such domains [3];

So, this paper specially designs the polysemantic semantic embedding module to extract the integrated semantic expressions of words; finally, the mixed feature word embeddings and polysemantic semantic word embeddings are fused and entered into the decoder through the mechanism of multi-head attention in order to realize more accurate entity extraction.

2 RELATED WOEK

The evolution of Named Entity Recognition (NER) in Cyber Threat Intelligence (CTI) can be divided into three distinct stages.

In the initial stage, most studies relied on hand-crafted rules to achieve satisfactory results. For instance, [4] examined the regular expression matching method and applied it to identify various types of attacks. This approach was subsequently utilized for deep packet inspection in tools such as Snort, 17-filter, and Bro. Additionally, bootstrapping algorithms have been employed as a complementary strategy to enhance the applicability and efficiency of rule-based matching methods. Rule-based models tend to perform effectively when the rules are comprehensive.

However, these approaches require significant manual effort and are challenging to adapt to new tasks or text types. To address the limitations of rule-based models, supervised machine learning methods—such as Support Vector Machines (SVMs), Hidden Markov Models (HMMs), and Conditional Random Fields (CRFs)—have been employed to automatically learn patterns from unstructured textual data.

In the intermediate phase, as deep learning evolves, the focus shifts to neural network models for Named Entity Recognition (NER) task in Cyber Threat Intelligence text (CTI). Zhou et al. [5] utilized Bi-LSTM-CRF architectures to tackle the cybersecurity NER task, subsequently integrating the extracted features with word-level vectors from the GLoVe model. Wu et al. [6] applied Bi-LSTM, CNN, and CRF architectures to the cybersecurity NER task. The primary distinction among these approaches lies in the use of the attention mechanism, which assigns corresponding weights to the extracted token features. Additionally, the label decoder captures the association information between the labels.

In the recent phase, Gao et al. [7] introduced a data- and knowledge-driven cybersecurity NER model. While these models demonstrated promising results, they overlooked the polysemous nature of entities in the cybersecurity domain, which complicates the semantics of intelligence data and ultimately results in inefficiencies in entity extraction.

3 PROPOSED MODEL

This section presents the details of the SEM model proposed in this paper, the overall architecture consists of four parts: Input layer, Representation layer, Attention layer, and Entity layer. Given a sentence S(X, Y) with n words in the cyber security domain, as

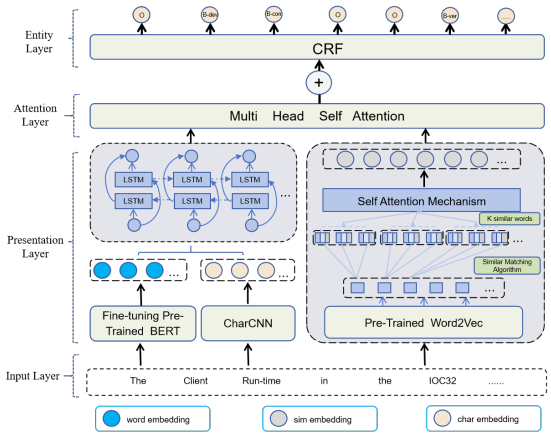

Figure 1: Architecture of SEM.

shown in Eq (1).

$$S(X,Y) = \begin{cases} X = [w_1, w_2, \ldots, [w_{k-1}, w_k, w_{k+1}], \ldots, w_n] \\ Y = [O\ ,\ O, \ldots,\ B\ ,\ I\ ,\ I,\ \ldots\ ,\ O] \end{cases} \quad (1)$$

In this paper, we consider NER as a sequence labeling task with a BIO labeling schema; w_i is each word in the sentence in X, and the elements in Y are the labels corresponding to each word, where a set of meaningful words is an entity, and w_{k-1}, w_k, w_{k+1} is an entity in Eq (1).Therefore, the goal of this paper is to learn a parameterized mapping $f_\theta : X \rightarrow Y$ from input words to output labels.

In this paper, we first input text into the BERT language model to obtain word-level word embeddings containing contextual representation information. Then, similar-meaning representations of words are constructed through the similar semantic embedding module. Subsequently, the mixed feature word embeddings similar-meaning word embeddings are combined through a multi-head attention mechanism to generate integrated semantic expressions. Finally, the results are passed to the decoder and CRF of the feed-forward neural network to obtain the set of labels Y corresponding to set X. The details of the SEM model as shown in Figure 1.

In this paper, we use BERT as the word embedding output model. Specifically, given a word $w_i (1 < i < n)$, this paper utilizes Eq (2) to obtain the corresponding word embedding vector. Where, $e_i^w \in R^{dw}$, d_w is the dimension of the word-level embedding vector:

$$e_i^w = \text{BERT}(w_i) \quad (2)$$

Where *BERT()* denotes the input w_i to compute the word level word embedding operation.

Since there is no index record for unknown words in the vocabulary of the pre-trained model, which makes it difficult to generate relevant word vectors, this paper introduces character-level features in the embedding layer, which is based on the hidden state of character-based convolutional neural networks, and research has demonstrated that the introduction of character-level features can help the model to better deal with unregistered words and spelling errors, as well as to capture morphological and structural information about the words, thus improving the model's performance in natural language processing tasks. Finally, by using Eq (3) and Eq (4), the word-level embeddings e_i^w and character-level

e_i^c embeddings are concatenated to obtain the final mixed feature word embedding representation O_i:

$$e_i^c = \text{CharCNN}\left(w_i\right) \tag{3}$$

$$O_i = \text{concat}\left(e_i^w, e_i^c\right) \tag{4}$$

Where, where *CharCNN()* denotes a character level word embedding operation computed on the input; *concat()* is a vector splicing operation.

Secondly, as shown in the ALGORITHM 1 ,we use the self-attention mechanism for S-SET computation to calculate the semantic relatedness weights $\emptyset_{ij}$ between the target word and similar words by Eq (5) and regularize it by Eq (6) , then accumulate it by Eq (7) to get the word embeddings with polysemantic semantics s_i. Finally,by Eq (8), we then enter *BiLSTM* layer to obtain word vectors with contextual information h_t^s:

$$\emptyset_{ij} = \text{cosine}\left(z_i * v_{ij}\right) \tag{5}$$

$$\partial_i = \frac{\exp(\emptyset_{ij})}{\sum_1^K \exp\left(\emptyset_{ik}\right)} \tag{6}$$

$$s_i = \sum_{i=1}^{K} \partial_i * z_i \tag{7}$$

$$h_t^s = \text{BiLSTM}\left(\vec{h}_{t+1}^s, \overleftarrow{h}_{t-1}^s, s_t\right) \tag{8}$$

where $\vec{h}_{i+1}^w$ is forward LSTM, $\overleftarrow{h}_{i-1}^w$ is backward LSTM and BiLSTM denotes bidirectional LSTM.

Subsequently, in this paper, mixed feature word embeddings and polysemantic semantic word embeddings are computed and fused between the embedding vectors at different levels through a multi-head attention mechanism to obtain H_t that is an integrated semantic representation with integrated semantics through Eq (9), and finally enter the CRF layer to predict the entity category:

$$H_t = \text{Multi Attention}\left(h_t^s, h_t^w\right) \tag{9}$$

Algorithm 1 Similar Matching Algorithm

Input: X and V are obtained from Word2Vec training, threshold M
Output: Filter result Z
Load Word2Vec model
For in X do
  Select the w_{ik} by most_similar(x_i)
// The embedded function of Word2Vec
W ={w_{i1}, w_{i2}, w_{i3}, . . ., w_{im}}
  Compute contrast similarity by using fuzzy string matching set Q
  If Q>M then
Put it to L
L = {w_{i1},w_{i2},w_{i3},...,w_{ik}}
Z = {v_{i1},v_{i2},v_{i3},...,v_{ik}}
Pass in to the next layer
end

4 EXPERIMENT

In this section, this paper first introduces the datasets and experimental results. Then, the benchmark modeling analysis on both datasets is given. In addition, an ablation study is performed. Finally, some insights for further analysis are given.

4.1 dataset

This paper utilizes experimental datasets from two sources. The first is DNRTI [8], a large-scale dataset for NER in CTI, comprising 6,574 annotated sentences and 36,412 entities. These entities are categorized into 13 groups, including attacks, sample files, and more.

The second dataset, provided by Bridges2 et al. [9], is collected from various platforms within the network security domain and contains multiple entities, such as applications, vendors, operating systems, and related terms.

4.2 model comparison

In this section, the proposed model is compared with several baseline models.

A detailed baseline is presented below.

Abdullah et al. [4] provided a Conditional Random Field (CRF) model, which is a statistically grounded conditional probability distribution model commonly employed for sequence labeling. The CRF is based on Markov random fields and utilizes maximum likelihood estimation to learn features.

Zhou et al. [5] employed Bi-LSTM and CRF architectures to carry out cybersecurity Named Entity Recognition (NER) tasks. Initially, the Bi-LSTM layer extracts contextual features from the input embeddings, followed by the CRF layer, which decodes the sequences and generates the output tokens.

Wu et al. [6] utilized Bi-LSTM, CNN, and CRF to execute the NER task. Compared to the aforementioned baselines, they employed a linear stack of LSTM and CNN in the deep neural network layer to achieve more efficient global and local feature representation. Liu et al. [10] introduced a semantic enhancement approach for the NER task, where the semantic information of words is primarily conveyed through soft and hard semantic enhancement techniques.

Gao et al. [7] proposed a data- and knowledge-driven Named Entity Recognition (NER) model for cybersecurity. At the input layer, an external dictionary serves as an auxiliary knowledge base to enhance word representation. Additionally, a self-attention mechanism is incorporated to capture the intrinsic dependencies between words in a sentence.

4.3 experimental results

In this paper, we select the comparison of the best work in recent years with the following models Abdullah [4], Zhou [5], Wu [6], Liu [10], Gao et al [7]. Table 1 and Table 2 present the experimental results of the model proposed in this paper and the baseline model across the two datasets. From the Table 1 and Table 2, it is evident that the SEM model achieves the best performance. For a dataset like DNRTI to test the SEM model, when word embeddings are embedded using the model's final integrated semantic word embeddings, the precision, recall, and f1-score reach 86.86, 85.96,

Table 1: Main comparison results of DNRTI dataset.

Methods	Precision	Recall	F1-score
CRF(CRFsuit)	85.85	73.06	78.94
BiLSTM+CRF[37]	74.07	75.98	75.01
IDCNN+CRF[45]	74.42	77.40	75.88
LSTM+BiLSTM+CRF	-	-	67.09
CNN+BiLSTM+CRF[37]	76.20	74.07	76.14
Liu et al.(2022)	86.16	84.54	85.34
$\text{SecureBERTfinetunning}_{\text{ours}}$	81.71	80.84	80.00
$\text{Bestmodel}_{\text{ours}}$	82.40	82.06	81.73
$\text{FinalModel}_{\text{ours}}$	**86.50**	**85.96**	**85.44**

Table 2: Main comparison results of the Bridges et al. Dataset.

Methods	Precision	Recall	F1-score
Abdullah et al. (2018)	80.76	76.27	78.45
Zhou et al.(2021)	85.16	80.70	83.37
CRF(CRFsuit)	85.25	84.23	83.24
LSTM-GCN-CRF	86.03	82.57	84.26
Wu et al.(2022)	85.69	83.58	84.62
Gao et al.(2021)	90.19	86.60	88.36
$\text{Basemodel}_{\text{ours}}$	89.71	89.11	88.53
$\text{FinalModel}_{\text{ours}}$	**93.12**	**92.56**	**92.02**

and 85.44, respectively, which are overall better than all existing baseline models..

Among the compared models, most of the models such as CNN+BiLSTM+CRF do not cope well with the polysemy of intelligence entities, and most of them obtain the polysemy of words through contextual information, but once encountering unknown or sparse words in the domain vocabulary, there is usually only a small portion of the contextual information as well as the complex contextual information that requires background knowledge to be utilized, and the contextual information is not enough to clearly distinguish the specific meaning of the entity, Similarly, the CRF model based on CRFsuite library is also. Therefore, SEM models tend to achieve higher precision, recall, and F1 scores compared to all baseline models, including those from Bridges et al. dataset, again the results obtained in this paper are the best results, and therefore, from a comprehensive point of view, the model in this paper performs the best in the above two datasets compared to all baselines.

4.4 model analysis

To verify the rationale behind the components of the SEM model, this section conducts several ablation experiments, with the results presented in Table 3.

Specifically, after removing the multiple attention module, this paper observes a slight decrease in the performance of the model. The F1 score shows a difference of 0.34 in the first dataset and 0.61 in the second dataset. This phenomenon demonstrates the effectiveness of multiple self-attention mechanisms in learning lexical interactions across different subspaces.

After removing the character word vector component, the three evaluation metrics decreased slightly, especially the precision. This suggests that character word vectors can help the model better handle unregistered words, spelling errors, while capturing morphological and structural information of words.

Similarly, when the integrated semantic word vector embedding component is removed, the evaluation metrics simultaneously drop sharply precision in the first dataset by a gap of 2.70, and in the second dataset by a gap of 2.19. This suggests that the polysemantic embedding module can indeed be used to extend the learning of word polysemantics from complex threat intelligence in terms of enhancing the efficiency of unknown entity recognition.

In summary, the substantial performance gap between the SEM model and the ablation model shown in Table 3 highlights the effectiveness of the various components.
where Att denotes a multi-head attention mechanism; Sim denotes a polysemantic semantic module; Char denotes a character-level word vector

5 CONCLUSION

In this research paper, we propose a new SEM model to improve the efficiency of entity recognition in CTI text. In the embedding layer, this paper adopts the traditional BERT to obtain word-level word embeddings with contextual information, and then character word embeddings are generated using character convolutional networks and fused with word-level word embeddings to generate more reasonable and effective word embedding vectors. In addition, this paper is to utilize the attention mechanism to construct a similar semantic space for keywords in threat intelligence through similar

Table 3: Ablation Experiment.

Methods	DNRTI			Bridges et al.		
	P	R	F1	P	R	F1
SEM	**86.50**	**85.94**	**85.44**	**94.62**	**93.30**	**92.02**
-Sim	84.16	83.59	83.02	92.43	91.77	91.12
-Att-Sim	82.98	82.54	82.11	91.23	90.16	89.12
-Sim-Char	83.11	82.66	82.23	90.89	90.52	90.17
-Att-Sim-Char	82.55	82.21	81.89	90.11	89.81	89.56
Base model	82.40	82.06	81.73	89.71	89.11	88.53

"-" and '+' indicate deletion and addition of relevant components

lexical meaning algorithm, so as to amplify the polysemous nature of cybersecurity entities, in order to solve the problem of difficult to recognize the polysemous nature of entities.

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